X3N-O-THON 2026

TEAM NAME : SYNAPSE_GTB

DOMAIN : Healthcare Track

CATEGORY : AI/ML-powered Web Application

PROJECT TITLE : Project Chronos

PROBLEM STATEMENT

User Pain Points

Hospitals generate massive amounts of physiological data (vital signs, labs, diagnoses), but most monitoring systems only **alert clinicians after a critical threshold is crossed** .

Doctors often experience:

- **Alarm fatigue** from too many reactive alerts
- Difficulty identifying **subtle deterioration patterns**
- Lack of **predictive insights** from continuous patient data

As a result, patient deterioration such as **sepsis, shock, or cardiac arrest** is often detected **too late** , reducing survival chances.

Technical / Operational Challenges

Current systems face several limitations:

1. **Reactive Monitoring**
   Traditional monitors trigger alerts only when vitals exceed predefined thresholds.
2. **Data Silos**
   Patient data is distributed across multiple hospital systems.
3. **Limited Predictive Intelligence**
   Most hospital systems do not use machine learning to detect early risk patterns.
4. **Cognitive Load on Clinicians**
   Physicians must manually interpret large volumes of patient data under time pressure.

Why This Problem Matters Now

Healthcare systems are becoming increasingly **data-rich but insight-poor** .

Recent trends driving urgency:

- Rising ICU and emergency department load
- Increasing clinician burnout
- Availability of large clinical datasets like **MIMIC-IV**
- Advances in **machine learning for healthcare**

AI can now detect **patterns invisible to human monitoring** , enabling earlier intervention.

PROPOSED SOLUTION

CORE IDEA / CONCEPT

Project Chronos is an AI-driven early warning system that analyzes patient vital signs and clinical data to **predict clinical deterioration hours before it occurs**.

Instead of reacting to crises, the system **anticipates risk** and alerts clinicians in advance.

KEY FEATURES

1. Early Risk Prediction

Machine learning models analyze patient vitals to estimate the probability of deterioration.

2. Continuous Monitoring

The system continuously processes incoming patient data and updates risk scores in real time.

3. Intelligent Alerting

Instead of threshold alarms, the system generates **predictive alerts** based on risk trajectories.

4. Clinical Dashboard

A visualization dashboard displays:

- patient risk scores
- vitals timeline
- predicted deterioration probability

5. Explainable AI

The system highlights **which vital signs contributed most to the prediction**, improving trust among clinicians.

HOW IT ADDRESSES THE CHALLENGES

Challenge	Solution
Reactive monitoring	Predictive machine learning models
Data overload	Automated risk scoring
Alarm fatigue	Intelligent alerts based on trends
Complex data interpretation	Visual dashboards

WHY THIS SOLUTION IS BETTER

Existing hospital monitoring systems mainly rely on **fixed thresholds**.

Example:

Alert if heart rate > 130

This allows clinicians to **intervene earlier**, potentially preventing life-threatening events.

Predict probability of deterioration in next 6 hours

THE INNOVATION

WHAT MAKES IT UNIQUE / ANTI-BORING

Three aspects make the solution stand out:

1. Predictive Healthcare

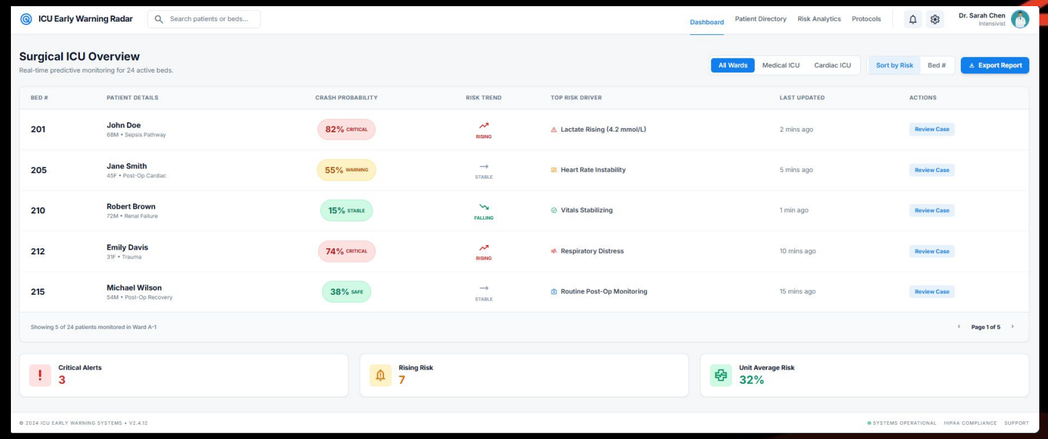
Moves from **reactive medicine → predictive medicine**

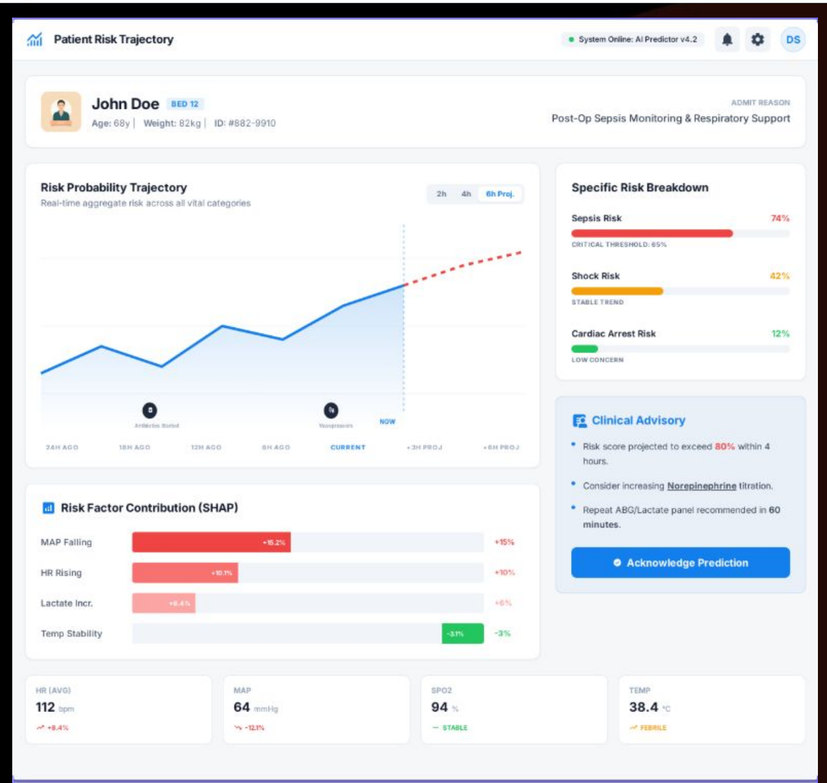
2. Real Clinical Data

The system is built using the **MIMIC-IV-ED dataset**, which contains real hospital patient records.

3. Scalable Architecture

The system is designed to integrate with **hospital electronic health record systems**.





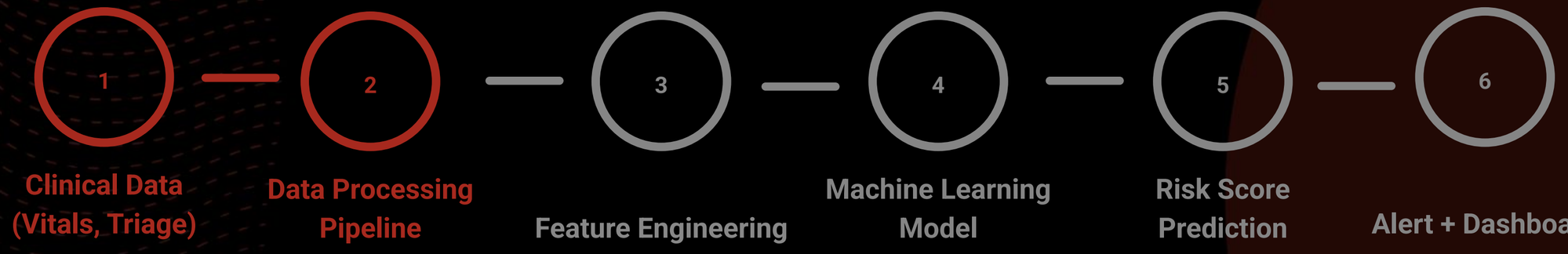
TECHNICAL DESCRIPTION

Category	Technology
Prog.Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn / XGBoost
Visualization	Streamlit
Database	PostgreSQL / BigQuery
Dataset	MIMIC-IV-ED Demo
Deployment	Docker / Cloud

System Components

1. **Data Ingestion Layer**
   Imports clinical data from MIMIC dataset.
2. **Feature Engineering Layer**
   Processes vital signs into ML features.
3. **Prediction Engine**
   Machine learning model predicts risk score.
4. **Alert System**
   Generates alerts if risk crosses threshold.
5. **Visualization Dashboard**
   Displays patient data and predictions.

DATA FLOW



IMPACT & BENEFITS

EXPECTED IMPACT

The system could significantly improve clinical decision-making by providing **earlier insights into patient deterioration** .

Potential impact areas:

- ICU monitoring
- emergency department triage
- remote patient monitoring

SHORT-TERM VS LONG-TERM VALUE

Short Term

- Prototype demonstrating predictive monitoring
- Decision support for clinicians

Long Term

- Integration into hospital monitoring systems
- Deployment in ICU and emergency departments
- Large-scale predictive healthcare systems

MEASURABLE IMPROVEMENTS

Potential measurable improvements include:

- earlier detection of deterioration
- reduced ICU mortality
- reduced emergency response time
- improved clinician decision support

Benefit	Explanation
Speed	Automated real-time risk predictions
Accuracy	Machine learning detects complex patterns
Simplicity	Easy-to-use clinical dashboard
Accessibility	Scalable cloud-based deployment
Early Intervention	Enables preventive treatment

FEASIBILITY & VIABILITY

TECHNICAL FEASIBILITY

The system is technically feasible due to:

- publicly available clinical datasets
- mature machine learning frameworks
- scalable cloud infrastructure

A prototype can be built using the **MIMIC-IV demo dataset** .

SCALABILITY & FUTURE ENHANCEMENTS

Future extensions may include:

- training on **full MIMIC-IV dataset**
- deep learning models (LSTM / Transformers)
- real-time hospital data integration
- wearable device monitoring
- personalized patient risk modeling

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